

Research paper

Long-tailed detection and classification of wafer defects from scanning electron microscope images robust to diverse image backgrounds and defect scales[☆]

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ABSTRACT

In semiconductor engineering, high yield of wafers relies on accurate detection and classification of wafer defects. The dataset for detecting wafer defects presents three primary challenges: (i) different background types, (ii) variable image or defect scales, and (iii) imbalanced data with a long-tailed distribution of defect types. These challenges create significant limitations for traditional classification techniques. To address these issues, we propose a stratified framework called Wafer Detection and Classification (WaferDC), designed specifically for detecting and classifying wafer defects from scanning electron microscope (SEM) images. Our framework achieves high defect detection performance on SEM wafer images by utilizing a multi-cluster memory bank, which effectively handles the challenges of (i) variable background types and (ii) differing image or defect scales. Building on this robust detection, we propose Segmentation and Mix (SegMix), a novel defect augmentation technique based on anomaly heatmaps, which enhances the reliability of defect detection and classification in a (iii) long-tailed imbalanced environment. Finally, we pass defect-classified images through a parameter-efficient fine-tuning (PEFT)-based classifier (Shiet et al., 2023) utilizing a vision transformer (ViT) architecture, further improving overall defect detection and classification performance. We rigorously tested WaferDC on a proprietary SEM wafer dataset and the public Describable Textures Dataset-Synthetic (DTD-Synthetic) and Magnetic Tile Defect (MTD) datasets. The results confirm the effectiveness of our method in improving defect detection and classification in wafer manufacturing. Our code is available at <https://github.com/SpatialAILab/WaferDC>.

1. Introduction

As the demand for various electronic products and the Artificial Intelligence (AI) industry continues to rise, the importance of semiconductor production technologies is becoming increasingly significant. Achieving high yields is crucial in semiconductor manufacturing, and a key factor of this is the accurate detection and classification of defects on wafers. Thus, it is essential to further subdivide critical defect classes, as new defect types continuously emerge. Although these defects may appear similar, their underlying causes—such as material

issues or equipment malfunctions—can differ significantly, making it crucial to identify and address each cause individually and efficiently. This research is necessary to enable faster, targeted responses to these specific defect origins, thereby maintaining high wafer yields and improving overall production quality and efficiency (Nakata et al., 2017; Nakazawa and Kulkarni, 2018).

Wafer defect inspection typically uses two methods: optical and electron beam (e-beam) inspection (Zhu et al., 2022a). Optical inspection covers a wide region of interest (RoI) quickly, but struggles to

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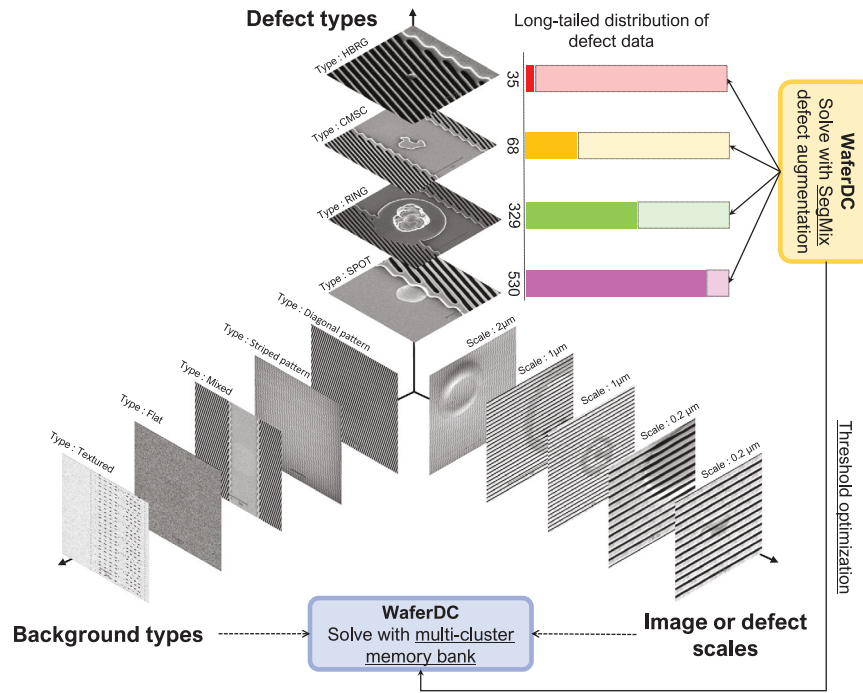


Fig. 1. An illustration of the three primary challenges associated with wafer defects, which include diverse background types, variable image or defect scales, and imbalanced data with a long-tail distribution of defect types. We resolve challenges related to varying backgrounds and scales with our newly proposed multi-cluster memory bank, while addressing the long-tailed distribution of defect data through our novel SegMix defect augmentation method.

detect small defects. In contrast, e-beam methods, such as Scanning Electron Microscope (SEM) imaging, provide high-resolution images of smaller areas, making them ideal for defect detection, but impractical for full wafer scanning. SEM images present challenges due to random defect locations, variable backgrounds, and image or defect scales. Our research aims to develop a framework that accurately detects and identifies small defects in SEM images to improve wafer yield. When constructing a wafer dataset based on SEM images, traditional classification methods face significant challenges due to the (i) different background types and (ii) variable image or defect scales inherent in SEM images, as well as the fact that (iii) some defects are rare, leading to imbalanced data with a long-tailed distribution of defect types as illustrated in Fig. 1. We empirically observe in Section 4 that these issues lead to suboptimal defect detection and classification accuracies with standard anomaly detection and classification algorithms, and consequently devising a method that can effectively address the challenges of (i), (ii), and (iii) to improve performance in both wafer defect detection and classification serves as our main motivation for this work.

In this paper, we propose a new framework called WaferDC that is tailored to address above challenges observed in wafer defect detection and classification SEM images. WaferDC achieves high detection performance on SEM wafer images by employing a multi-cluster memory bank strategy to address diverse image backgrounds and variable image or defect scales, allowing the defect detection process to be tailored for specific textures. Building on these enhanced detection results, we propose SegMix, an anomaly heatmap-based augmentation technique. Images generated from SegMix are further utilized for threshold optimization, aligning with real-world industrial settings and strengthens classification accuracy in environments with long-tailed, imbalanced data distributions. Lastly, we employ a PEFT-based classifier (Shi et al., 2023) with a vision transformer (ViT) architecture to achieve strong overall classification performance. Our contributions are summarized below:

- + We introduce WaferDC, designed to address the three challenges mentioned above and enhance wafer defect detection and classification accuracies in SEM images.

- + To improve the performance of wafer defect detection across various SEM image backgrounds and scales, we propose a novel multi-cluster memory bank strategy, where each cluster represents a different background type and scale in wafer defects. This also contributes to the enhanced quality of augmented defect images later on.
- + We introduce a new defect augmentation method called SegMix, which utilizes the anomaly segmentation map from the multi-cluster memory bank to extract precise defect areas and adopts smooth blending to synthesize realistic SEM images with defects. As will be shown later, the added synthetic data is not just confined to train the defect classifier but also used to adjust defect detection thresholds for the multi-cluster memory bank, leading to an overall enhanced framework performance.
- + We have extensively evaluated WaferDC on a proprietary SEM wafer dataset and public DTD-Synthetic (Aota et al., 2023) and the Magnetic Tile Defect (MTD) (Huang et al., 2020) datasets. Our experimental results demonstrate the capability of our approach in enhancing defect detection and classification performance.

The remaining of this article is organized as follows: We provide a literature review of relevant studies in Section 2. Our proposed method is illustrated in Section 3. We report and discuss experimental results in Section 4. The conclusion and limitations of this work can be found in Section 5.

2. Related work

In this section, we describe related work on wafer defect inspection, image defect detection and long-tailed classification related to our proposed method.

2.1. Wafer defect inspection

In recent years, efforts are being made to detect defects in wafer chips with the aim of improving the production yield of semiconductors (Nakata et al., 2017; Nakazawa and Kulkarni, 2018; Ooi et al.,

2013). The defect inspection methods for wafers can be categorized into two groups, namely optical and e-beam approaches (Zhu et al., 2022a). The optical method rapidly captures low-resolution images of large wafer dies to assess the presence of defects across the entire wafer, creating wafer maps that help identify defect patterns. Recently, many deep learning-based methods have been employed to enhance defect detection using wafer maps (Choi and Suh, 2025; Bae and Kang, 2023; Yu et al., 2021; Yu and Liu, 2020; Alawieh et al., 2020; Tsai and Lee, 2020; Yang and Sun, 2022). However, a key limitation lies in the difficulty of identifying the broad underlying rules that govern defect patterns, making it challenging to precisely distinguish individual defect types and shapes. This issue creates a critical weakness by limiting effective root cause analysis. Efforts have also been made to improve defect detection by applying deep learning techniques to images from real-world automated optical inspection (AOI) systems, which use optical cameras to examine more detailed areas (Cheng et al., 2023). While this approach enables a finer optical inspection, it has an inherent limitation in that the resolution of the captured images remains insufficient for accurate classification.

Another approach, the electron beam method, typically involves using high-resolution techniques like SEM imaging to capture localized areas of the wafer and identify defects. In contrast to the limitations of the optical approach, the electron beam method typically uses high-resolution techniques like SEM imaging to capture finer, localized areas of the wafer. This enables precise defect classification, allowing for faster root cause analysis, which in turn can help increase wafer yield (Zhu et al., 2022a). In the context of semiconductor SEM image analysis, our study aims to classify defect types in greater detail, allowing for more accurate identification and rapid response to defect causes. This approach is intended to facilitate effective causal analysis and improve wafer yield.

2.2. Image-based defect detection

In exploring the realm of defect detection (Liu et al., 2024), supervised methodologies (Venkataramanan et al., 2020; Ding et al., 2022; Chu and Kitani, 2020), one-class classification (OCC) methods (Roth et al., 2022; Defard et al., 2021; Bae et al., 2023; Liu et al., 2023; Lu et al., 2023b; Cohen and Hoshen, 2020a; Bergmann et al., 2020), and more recently, few-shot defect detection have been utilized (Jeong et al., 2023; Xie et al., 2023; Fang et al., 2023). Our work is based on the OCC setting centered around the concept of learning representations from clean normal data, which is a widely adopted practical setting for industrial applications. Since OCC methods require data from only one class, they are ideal when data from other classes (like anomalies or defects) is rare, expensive, or difficult to obtain. In this OCC setting, there are two main types of methods, namely feature embedding-based methods and reconstruction-based approaches. The main methods for feature embedding-based approaches are based on memory banks (Roth et al., 2022; Cohen and Hoshen, 2020b; Bae et al., 2023), which stores features from normal images in a memory bank and utilizes the distance between the query image features and the stored features in the memory bank to classify anomalies. Reconstruction-based anomaly detection methods (Shin et al., 2023b; Zavrtanik et al., 2021; Lu et al., 2023a; Yang et al., 2023, 2019; Qian et al., 2021) aim to reconstruct normal patterns, identifying deviations from these patterns as anomalies. However, managing the reconstruction capability of the network remains challenging as these models produce plausible reconstruction results even for abnormal inputs.

Handling multi-background and multi-scale anomalies. Several efforts have been made to address multi-background and multi-scale anomalies based on reconstruction-based approaches. Yang et al. (2023) proposed a multi-category decomposition editing network (MC-DEN) for texture defects, relying heavily on predefined libraries containing texture categories, which introduces limitations in handling unseen defect types. Yang et al. (2019) and Qian et al. (2021) both

proposed a clustering layer that computes the target distribution based on the probability of assigning each sample to a cluster for defect detection. This approach relies on the target distribution to minimize KL-divergence loss between the soft assignment and target distributions, which increases computational complexity due to additional distribution calculations. In addition to increased training time, reconstruction-based methods yield poorer detection accuracies compared to feature embedding-based approaches, making them insufficient to meet the demands of industrial manufacturing (Defard et al., 2021; Liu et al., 2023; Lv et al., 2023; Perera et al., 2019).

Other method includes the work of Campos-Romero et al. (2024) (InAI), which employs a multi-scale approach using multiple extraction points within convolutional neural network layers to incorporate varying levels of resolutions and receptive fields. This framework follows a distribution-based approach leveraging normalizing flow (Rezende and Mohamed, 2015), which transforms the distribution from the base space to a target latent Gaussian space. However, it may fail to learn a meaningful distribution as it emphasizes local pixel correlations rather than capturing semantic and spatial information across multi-scale feature maps (Kirichenko et al., 2020). Moreover, the loss of information arises from the averaging of negative log-likelihoods during the calculation of anomaly scores (Rudolph et al., 2021). For these reasons, methods based on normalizing flows exhibit certain limitations in anomaly detection.

To the best of our knowledge, our work marks one of the first efforts to extend the lead enjoyed by the feature embedding-based methods for multi-background and multi-scale anomalies, demonstrating improved performance as shown in Section 4.

2.3. Long-tailed image classification

Traditional approaches for long-tailed classification involve re-sampling (Chang et al., 2021) imbalanced samples during training or modifying loss functions (Ren et al., 2020; Du and Wu, 2023; Hong et al., 2021; Cao et al., 2019) to address class imbalance. Other methods adjust model outputs (Menon et al., 2021; Tao et al., 2023) or use ensembling techniques (Wang et al., 2021; Zhang et al., 2022) trained on balanced subsets. Self-supervised learning approaches (Zhu et al., 2022b; Park et al., 2022; Du et al., 2023) have also been used to enhance representation learning in limited data scenarios.

Recently, pretrained large-vision models like ViT (Dosovitskiy et al., 2020), Contrastive Language-Image Pretraining (CLIP) (Radford et al., 2021), and Self-distillation with no labels (DINO) (Caron et al., 2021) have gained attention for long-tailed classification (Dong et al., 2022; Ma et al., 2021). To prevent overfitting on smaller long-tailed datasets, parameter-efficient fine-tuning (PEFT) has been applied, with Parameter-Efficient Long-Tailed Recognition (PELT) (Shi et al., 2023) showing its effectiveness through logit-adjustment loss and test-time ensembling.

Shin et al. (2023a) emphasizes that adding more samples to long-tailed data outperforms other state-of-the-art methods for long-tailed recognition, necessitating synthesis of high-quality anomaly images for industrial anomaly detection. Thus, our approach focuses on defect augmentation to address long-tail challenges, enhancing both the fine-tuning of pretrained models and the training of models from scratch. By generating realistic defect data, WaferDC is optimized for real-world industrial long-tailed datasets, distinguishing it from prior work validated on toy datasets such as Canadian Institute for Advanced Research-10-Long Tail (CIFAR-10-LT) (Krizhevsky, 2009) and A large-scale hierarchical image database-Long Tail (ImageNet-LT) (Russakovsky et al., 2015).

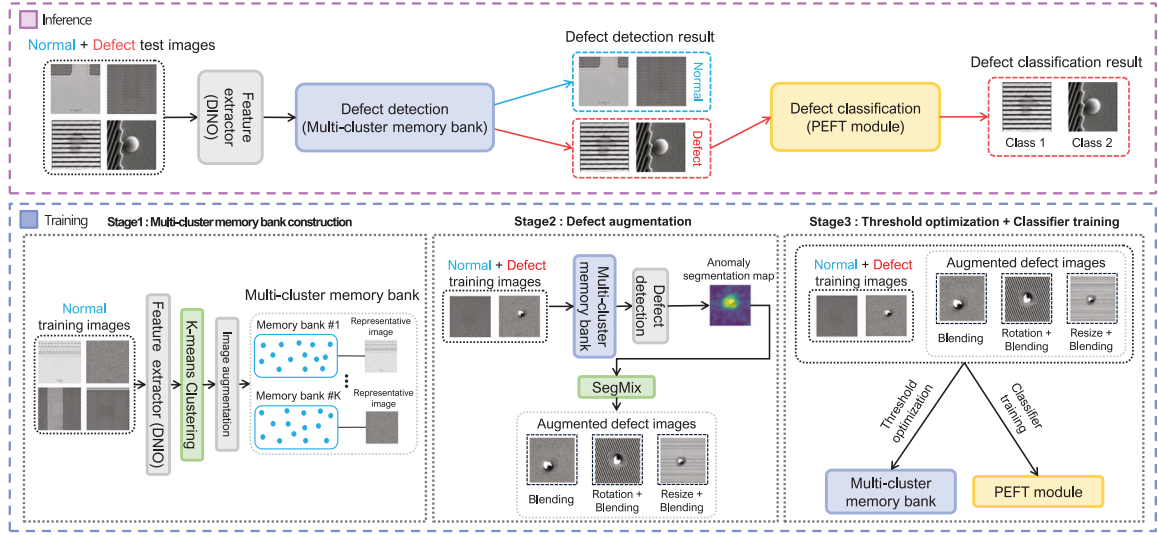


Fig. 2. An overview of our WaferDC framework includes both the inference and training processes. During inference, a pre-trained DINO feature extractor is employed to extract features, which then enables defect detection through the utilization of a multi-cluster memory bank. Detected defects are then classified using the PEFT module. During the training phase, our method involves three stages: The first stage uses the k -means clustering to construct the multi-cluster memory bank according to the texture of normal training images. In the second stage, SegMix generates augmented defect images based on the segmentation maps of the training defect images outputted from the multi-cluster memory bank. The third stage involves threshold optimization for the multi-cluster memory bank and conducting balanced learning with the PEFT module using the original and augmented defect images.

3. Proposed method

We now illustrate our WaferDC method for wafer defect detection and classification (see Fig. 2).

Problem setting. We assume that we are provided with a dataset that presenting three aforementioned challenges (i), (ii) and (iii). These conditions contribute to the complexity of accurately detecting and classifying defects in semiconductor manufacturing.

Preliminaries. We define the normal and defect training sets as $D^n := \{I_1^n, I_2^n, \dots, I_i^n\}$ and D^d respectively, where $I_i^n \in \mathbb{R}^{H \times W \times 3}$ and $I^d \in \mathbb{R}^{H \times W \times 3}$ are the normal and defect images respectively and (H, W) represents the height and width of input image respectively. i represents the image index in the normal set. We also define the test image as $I^t \in \mathbb{R}^{H \times W \times 3}$.

Inference overview. During the inference phase, WaferDC operates with two stratified modules: a defect detection module that utilizes a multi-cluster memory bank with a pretrained DINO (Caron et al., 2021) feature extractor and a defect classification module that employs a PEFT-based vision transformer (ViT) encoder with a cosine classifier (Shi et al., 2023). Initially, the detection module identifies defect images from normal images, and then, only the detected defects images are processed further by the classification module for classification of the type of defect.

Training overview. The training phase of WaferDC consists of three stages designed to optimize the functionality of both the detection and classification modules:

(i) **Multi-cluster memory bank construction:** We organize the data into a multi-cluster memory bank using k -means clustering and normal augmentation. This approach allows us to tailor the defect detection process to the specific textures in the data, resulting in more precise defect segmentation.

(ii) **Defect augmentation using SegMix:** Mitigating the insufficiency of defect data and its long-tailed distribution, we employ the SegMix method for enhanced defect augmentation. Originating from the results of our multi-cluster memory bank, SegMix uses anomaly heatmap (segmentation map) to precisely integrate defect regions with normal images, applying transformations like rotation and resizing to enhance the realism and diversity of the augmented data.

(iii) **Threshold optimization + classifier training:** After generating augmented images with SegMix, we incorporate these images into the validation set for each memory bank, enabling accurate defect detection optimal F1 threshold setting without relying solely on AUROC metrics from test data (Roth et al., 2022; Defard et al., 2021; Bae et al., 2023; Cohen and Hoshen, 2020b; You et al., 2022). This method simulates a real-world scenario in which the test set remains unseen, ensuring that our optimal F1 threshold is realistically calibrated using only the validation set. Additionally, the augmented images improve classifier performance by effectively balancing the long-tailed distribution of defect training images, leading to a more robust classification process.

3.1. Multi-cluster memory bank

Inspired by the success of OCC approaches (Roth et al., 2022; Defard et al., 2021; Bae et al., 2023; You et al., 2022; Lu et al., 2023b), which excel in environments rich in normal images, we aim to extend their strengths to handle diverse backgrounds and variable scale anomalies using a multi-cluster memory bank strategy. Traditional single memory banks (Roth et al., 2022; Cohen and Hoshen, 2020b) may not effectively handle the diverse backgrounds present in defect images. To overcome this limitation, we introduce a multi-cluster memory bank. This method allows for the categorization of features according to similar backgrounds and textures, enhancing defect detection and segmentation performance by enabling more accurate feature comparisons. These experimental results will be shown in Section 4.

Feature extraction. We perform feature extraction using the pre-trained DINO (Caron et al., 2021) model for all images, without fine-tuning it on the specific datasets. The feature extracted from the normal training image I_i^n is defined as $F_i^n \in \mathbb{R}^{784 \times 1536}$, in which the transpose of the p th row vector is the local patch feature $\mathbf{f}_{i,p}^n \in \mathbb{R}^{1536}$ of the p th patch out of 28×28 equally sized image patches. We also define $\mathbf{f}_i^n := [\mathbf{f}_{i,1}^n, \dots, \mathbf{f}_{i,784}^n]^T \in \mathbb{R}^{1204224}$, which is essentially the row-vectorized form of F_i^n . Similarly, the features extracted from the training defect image I^d and test image I^t are denoted as $F^d \in \mathbb{R}^{784 \times 1536}$ and $F^t \in \mathbb{R}^{784 \times 1536}$ respectively, and their corresponding row-vectorized forms are denoted as $\mathbf{f}^d$ and $\mathbf{f}^t$.

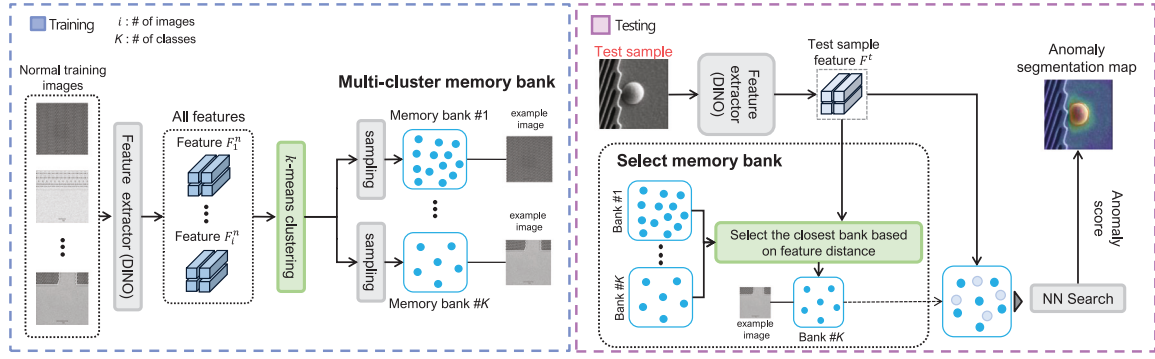


Fig. 3. An overview of our multi-cluster memory bank strategy. For training, we extract pre-trained DINO (Caron et al., 2021) feature F_i^n and vectorize it to yield an image feature f_i^n . To facilitate efficient clustering, principal component analysis (PCA) was applied to all feature vectors $\{f_i^n\}$. We then perform k -means clustering on the reduced image features to assign a bank ID for each normal image, applying random-crop augmentation to enhance robustness to scale variations. The multi-cluster memory bank is then constructed by sampling the original patch features $\{f_{i,p}^n\}$ of images assigned to each bank. During construction, we also yield a representative image feature r_k for each bank which is later used for assigning bank ID during the test phase. (An example image from each bank is displayed in the figure above.) During the testing phase, test image feature f^t is assigned the bank ID whose representative feature r_k is the nearest to f^t in terms of the L_2 -norm distance. Then, by utilizing the nearest neighbor search across all patch features of the test image I^t , $\{f_q^t\}$, we calculate the global anomaly score s of the test image for defect detection as well as yield the defect segmentation map S for defect localization.

Bank assignment. We employ the k -means clustering to assign the bank identifier (ID) for each training image I_i^n followed by construction of individual memory banks for each cluster.

For efficient clustering, we perform the principal component analysis (PCA) (Karl Pearson, 1901) with all long feature vectors $\{f_i^n\}$ of the normal training images $\{I_i^n\}$ to reduce the dimension of each feature vectors to 100. As a result, we yield a reduced feature $f_i^{n, \text{reduced}} \in \mathbb{R}^{100}$ for each normal training image I_i^n . We then use the set of reduced features $\{f_i^{n, \text{reduced}}\}$ from normal training images to perform k -means clustering.

Subsequently, we assign each training image I_i^n to one of K clusters by choosing the cluster whose centroid is closest to the reduced feature of the current image ($f_i^{n, \text{reduced}}$). Accordingly, the original high-dimensional feature vectors extracted by DINO (Caron et al., 2021) are stored in the multi-cluster memory bank. Specifically, once clustering is performed on the PCA-reduced features, the corresponding original DINO patch features of each image are grouped and stored based on their assigned cluster IDs. Thus, the output of the clustering process consists of the original image features organized by cluster.

We will assume for now that K is fixed, and illustrate the procedure for choosing K at the end of this subsection. In terms of equation,

$$C_k(I_i^n) = \arg \min_{k \in \{1, \dots, K\}} \|f_i^{n, \text{reduced}} - \mu_k\|_2^2, \quad (1)$$

where $C_k(I_i^n)$ denotes the assigned cluster of the i th training image i and μ_k is the centroid of the k th cluster. Initial centroids $\{\mu_k\}$ are randomly drawn from the pool of reduced features $\{f_i^{n, \text{reduced}}\}$ as for the standard k -means clustering procedure.

After the clusters are assigned, the reduced features $f_i^{n, \text{reduced}}$ are used to recalculate each centroid as follows

$$\mu_k = \frac{1}{|B_k|} \sum_{i \in B_k} f_i^{n, \text{reduced}}, \quad (2)$$

where B_k denotes the indices of image features in cluster k . Alternatingly solving for Eqs. (1) and (2) across all clusters yield the final bank IDs assigned to all normal training images later used for the construction of our multi-cluster memory bank strategy.

In our work, the optimal number of clusters (K) is determined by selecting the configuration that yields the highest silhouette score (Pedregosa et al., 2011), h , defined as

$$h = \frac{b - a}{\max(a, b)}, \quad (3)$$

where a is the average distance within a cluster and b is the minimum mean distance to the nearest different cluster. h essentially evaluates

Algorithm 1 Multi-cluster memory bank

Require: Training normal images $\{I_i^n\}$, Test images $\{I^t\}$, Number of clusters K

- 1: **Bank assignment:**
- 2: **for** each normal training image I_i^n **do**
- 3: Extract f_i^n then Perform PCA to yield $f_i^{n, \text{reduced}}$.
- 4: **end for**
- 5: Perform k -means clustering using Eq. (1).
- 6: Assign bank IDs to all normal training images.
- 7: **Bank construction:**
- 8: **for** $k = 1, \dots, K$ **do**
- 9: Add random-cropped augmentations of images
- 10: Construct bank M_k using Eq. (4).
- 11: Calculate representative feature r_k using Eq. (5).
- 12: **end for**
- 13: **Test phase:**
- 14: **for** each test image I^t **do**
- 15: Extract f^t and allocate using Eq. (6).
- 16: Calculate image-level anomaly score s using Eq. (10) and yield defect segmentation map S via upscaling, interpolation and smoothing.
- 17: **end for**
- 18: **return** s , S for each test image I^t

the degree of cluster separation, with higher scores signifying well separated clusters.

Bank construction. After assigning bank IDs based on the clustering results, we augmented the dataset by applying a normal augmentation technique involving random cropping of normal images, adding these augmented images to enhance robustness to scale variations. Following this augmentation, 10% of the images are designated as the validation set, while the remaining patch-level features are allocated to their corresponding memory banks. For each cluster k , we sample 1% of the patch-level features (for efficiency as in Roth et al., 2022) whose image features belong to cluster k , i.e. $\{f_{i,p}^n \mid i \in B_k\}$, to form the k th class memory bank M_k . With abuse of notations,

$$M_k = \bigcup_{i \in B_k} \text{sample}(\{f_{i,p}^n\}, 0.01), \quad (4)$$

where $\text{sample}(X, z)$ returns the set X with probability of z or otherwise returns an empty set.

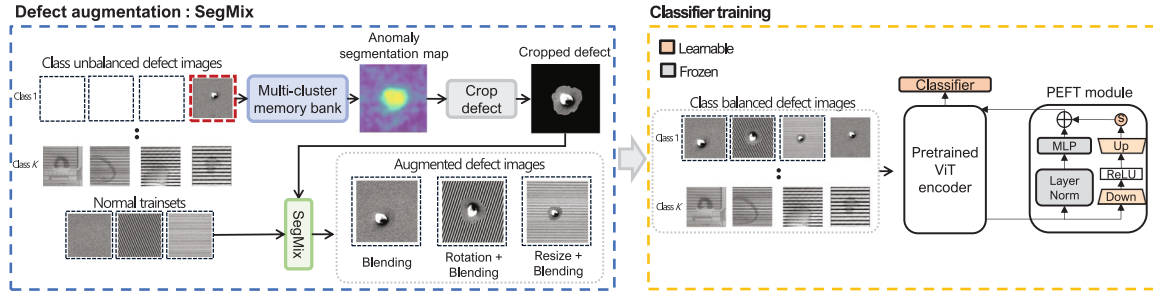


Fig. 4. An illustration of the SegMix framework. For class imbalanced defect datasets, we fill up the insufficient number of defect images for tail classes (illustrated as white black boxes from above) via augmentation with the defect detection module. Utilizing the defect segmentation map calculated through the same test phase in Fig. 3, we segment defect images into three distinct regions and SegMix blends the anomaly image with the normal image based on the mask. With class balanced defect datasets via augmentation, we train the classification module with the cross entropy loss. The classification module is composed of a pretrained ViT (Dosovitskiy et al., 2020) encoder and a single-layer cosine classifier, featuring a PEFT module that only trains a subset of parameters in each layer of the encoder. PEFT module incorporates the Adaptorformer (Chen et al., 2022), which adds a single learnable bottleneck layer in parallel to the feedforward network.

After the multi-cluster memory bank sets are formed, we finally yield the representative feature of each bank ID k , defined as $\mathbf{r}_k$, by aggregating the image features within the same class, i.e.

$$\mathbf{r}_k = \frac{1}{|B_k|} \sum_{i \in B_k} \mathbf{f}_i^n. \quad (5)$$

This representative feature is later used for allocating the test image to the class with the most similar background.

Test phase. During testing, we use the image feature $\mathbf{f}^t$ extracted from the testing image I^t to allocate the bank ID with the most similar background. In terms of equation, we allocate the bank ID k^* defined as

$$k^* = \arg \min_{k \in \{1, \dots, K\}} \|\mathbf{f}^t - \mathbf{r}_k\|_2. \quad (6)$$

After the bank ID is allocated, we apply the nearest neighbor (NN) search method in Roth et al. (2022) to calculate the anomaly score for each patch-level feature ($\mathbf{f}_q^t$ of I^t) which we define as s_q . For this purpose, we find the closest feature inside the memory bank M_k defined as $\mathbf{f}_{i^*,p}^n$, and calculate the distance between $\mathbf{f}_q^t$ and $\mathbf{f}_{i^*,p}^n$ to yield an unnormalized patch-level anomaly score s_q^{unnorm} i.e.

$$\mathbf{f}_{i^*,p}^n = \arg \min_{\mathbf{f}_{i,p}^n \in M_{k^*}} \|\mathbf{f}_q^t - \mathbf{f}_{i,p}^n\|_2, \quad (7)$$

$$s_q^{\text{unnorm}} = \|\mathbf{f}_q^t - \mathbf{f}_{i^*,p}^n\|_2 \quad (8)$$

For each patch q , the unnormalized anomaly score s_q^{unnorm} is normalized to s_q via exponential weighting function (Defard et al., 2021; Roth et al., 2022) of pairwise distances with patch-level features in the memory bank whose set is defined as B_k , i.e.

$$s_q = \left(1 - \frac{\exp(\|\mathbf{f}_q^t - \mathbf{f}_{i^*,p}^n\|_2)}{\sum_{\mathbf{f}_{i,p}^n \in M_{k^*}} \exp(\|\mathbf{f}_q^t - \mathbf{f}_{i,p}^n\|_2)} \right) s_q^{\text{unnorm}} \quad (9)$$

The final image-level (global) anomaly score used to identify anomaly images is defined as s and computed by finding the maximum patch-level anomaly score as follows:

$$s = \max_q s_q \quad (10)$$

Finally, we generate the pixel-wise defect segmentation map S by following the steps used in the anomaly detection literature (Defard et al., 2021; Roth et al., 2022). In summary, this is carried out by upscaling patch-level anomaly scores to match the image resolution, performing bilinear interpolation followed by Gaussian smoothing. The process of constructing and testing multi-cluster memory bank are summarized in Algorithm 1 (see Fig. 4).

3.2. Defect augmentation using SegMix

We propose a new defect-image augmentation method called SegMix, which crops the defective regions of defect images and blends to normal images with similar backgrounds without requiring any ground truth data (anomaly map) or manual labeling. While SegMix stems from CutMix (Yun et al., 2019), SegMix majorly differs from CutMix in that it avoids random cut-paste strategy and instead uses a continuous mask derived from the defect (anomaly) segmentation map to precisely crop the defect regions to create high-quality defect images.

Allocation of bank IDs for real defect images. To generate augmented images, we first allocate each defect training image to the closest memory bank by calculating the minimum L_2 -distance between the defect image feature $\mathbf{f}^d$ and the representative feature $\mathbf{r}_k$.

$$k^* = \arg \min_{k \in \{1, \dots, K\}} \|\mathbf{f}^d - \mathbf{r}_k\|_2 \quad (11)$$

After allocation, each defect training image yields an individual defect segmentation map S for the image as described in Section 3.1, Eqs. (7) and (9).

Generation of defect mask. Based on the patch scores s_q derived from the previous steps, we perform a rearrangement to match the image dimensions, creating the segmentation map S . Using the defect segmentation map S , we then build a continuous defect mask with the aim of seamlessly blending the anomaly regions with the normal images. Specifically, we assign the mask value of 1 for the pixels with anomaly scores greater than the average of minimum and maximum anomaly scores in S , which are defined as $s_{\min} := \min_{m,n} S_{(m,n)}$ and $s_{\max} := \max_{m,n} S_{(m,n)}$ respectively, where $S_{(m,n)}$ denotes the anomaly score of the (m,n) th pixel in S . We also assign 0s to the pixels whose anomaly scores are substantially lower than the average of min and max scores. All other pixels are treated as intermediate regions, assigning the value of 0.1 times the anomaly score of each corresponding pixel with the aim of providing smooth transition between the defect and normal regions. In terms of equations, we generate the 2D defect mask M whose (m,n) th entry ($M_{(m,n)}$) is obtained as

$$M_{(m,n)} = \begin{cases} 1 & s_{(m,n)} > \frac{s_{\min} + s_{\max}}{2} \\ 0 & 0 < s_{(m,n)} < \frac{s_{\min} + s_{\max}}{2} - 3\sigma \\ 0.1s_{(m,n)} & \text{otherwise,} \end{cases} \quad (12)$$

where σ is the standard deviation of anomaly scores in S . The thresholds are heuristically determined based on the observation that the normal pixels exhibit very small values close to 0 while the defect regions exhibit varying degree of anomaly scores. We then segment the defective region of the defect image to yield I_{def}^d defined as

$$I_{\text{def}}^d = M \odot I^d, \quad (13)$$

Algorithm 2 SegMix for defect image synthesis**Require:** Defect training set D^d , Normal training set D^n

```

1: Generated defects  $D^g = \{\}$ 
2: for each defect training image  $I^d$  do
3:   Extract  $\mathbf{f}^d$ 
4:   Allocate nearest memory-bank using Eq. (11)
5: end for
6: for each allocated training defect data  $I^d$  do
7:   Generate segmentation map  $S$  using Eq. (7) and Eq. (9).
8:   Define continuous mask  $M$  of  $I^d$  using Eq. (12).
9:   Segment defect region  $I_{def}^d = M \odot I^d$ .
10:  With probability of 1/3:
11:    Transform  $I_{def}^d$  using Eq. (14) to yield  $\tilde{I}_{def}^d, \tilde{M}$ .
12:  Draw a random normal image  $I_j^n$  from the same allocated
    memory bank.
13:  Apply Eq. (15) to synthesize defect image  $\tilde{I}^d$ .
14:   $D^g \leftarrow D^g \cup \tilde{I}^d$ 
15: end for
16: return Merge the defect training data,  $D^d \leftarrow D^g \cup D^d$ .

```

where $\odot$ denotes element-wise multiplication.

For generating diverse augmented defect images, we probabilistically apply rotation or resizing to I_{def}^d each with probability of 1/3. The angles for rotation were randomly chosen from multiples of 90 and 75 degrees, and the scaling factors for resizing were selected within the range of 0.5 to 1.0. In terms of equations, we yield the augmented defect region $\tilde{I}_{def}^d$ defined as

$$\tilde{I}_{def}^d = \begin{cases} \text{rotate}(I_{def}^d) & 0 < z < 1/3 \\ \text{resize}(I_{def}^d) & 1/3 < z < 2/3 \\ I_{def}^d & \text{otherwise,} \end{cases} \quad (14)$$

where $z \sim U(0, 1)$ is a uniformly distributed random variable between 0 and 1. Note that the mask M also undergoes the same transformation to yield $\tilde{M}$.

Synthesis of defect images. Generating an augmented defect image involves blending the masked defect region to a normal image I_i^n which belong to the same class in the multi-cluster memory bank as the ID allocated for the defect image I^d . In terms of equation, the augmented defect image $\tilde{I}^d$ is

$$\tilde{I}^d = \tilde{I}_{def}^d + (1 - \tilde{M}) \odot I_i^n. \quad (15)$$

Finally, we merge the real defect training set D^d with the generated defects D^g to create a class-balanced augmented training defect dataset. Details of this augmentation strategy are outlined in Algorithm 2.

Connections to vicinal risk minimization and mixup-based theory. SegMix instantiates the principle of Vicinal Risk Minimization (VRM) (Chapelle et al., 2000) for defect augmentation by constructing vicinal samples, which share conceptual similarity with VRM. Moreover, SegMix is similar to mixup-based theory (Zhang et al., 2018) as it blends defect regions of anomaly images onto normal images through interpolation. SegMix differs slightly by applying three distinct degrees of interpolation for normal, abnormal, and ambiguous regions, whereas mixup samples interpolation weights from a Beta distribution.

3.3. Threshold optimization and classifier training

The augmented images using SegMix are then utilized to optimize our model performance in two ways, (i) threshold optimization for defect detection and (ii) balanced classifier training for defect classification. The goal of threshold optimization is to enhance the model's ability to reduce the number of false positives and false negatives at

test time. The F1-optimal threshold is commonly used in the anomaly detection literature (Roth et al., 2022; Bae et al., 2023; Liu et al., 2023), to determine the decision boundary that balances precision and recall, which is formulated as the harmonic mean of the two metrics:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (16)$$

The optimal threshold is obtained by sweeping across a range of candidate values applied to the continuous anomaly scores predicted for each test sample. Beginning from a small positive value (e.g., 0.001), the algorithm incrementally adjusts the threshold and, at each step, computes the corresponding precision and recall by assigning samples with scores above the threshold as anomalous. We emphasize that the importance of optimizing the defect threshold is often overlooked in the aforementioned literature, which either report the AUROC that only focuses on the separability of anomaly samples and normal samples, or only show the (oracle) misclassification rate after determining the optimal threshold on the test samples (Roth et al., 2022). Contrary to these studies, we set validation thresholds during training and applied them during testing to simulate realistic industrial settings, ensuring a fairer comparison. Given the scarcity of defect images, we balance each bank's validation set by augmenting defect training images to match the number of normal images. This approach ensures accurate optimal F1 threshold calculation solely based on validation data.

For classification module we have followed PEL (Shi et al., 2023) with pre-trained ViT (Dosovitskiy et al., 2020) encoder for training and testing defect classification module. Both pre-trained backbones are trained on ImageNet-21K (Russakovsky et al., 2015). To ensure balanced classifier training, the process of generating augmented data is structured as follows: For a dataset with 10 defect classes, where the most prevalent class (head class) has 100 training images and the least prevalent (tail class) has 10, we aim to balance the distribution by creating 90 additional images. These images are augmented in accordance with the proportions allocated to each class in the multi-cluster memory bank.

Practical remark. Considering the application of our framework in real-world environments, frequent retraining is not necessary. As the production line stabilizes, the emergence of new defect types typically decreases to one to three per quarter, and in mature phases, new patterns may not appear for entire quarters.

4. Experimental results

In this section, we outline the experimental methodologies used to demonstrate the advantages of our WaferDC framework and illustrate the datasets used for our evaluations.

4.1. Datasets

For thorough evaluation of our proposed method, we compared WaferDC against other baseline and state-of-the-art approaches on three defect classification datasets, namely the SEM wafer dataset (private), the DTD-Synthetic (public) (Aota et al., 2023), and the MTD dataset (public) (Huang et al., 2020). The details of these datasets can be found in Table 1.

SEM wafer dataset. In our study, we evaluated the proposed method using our in-house SEM wafer dataset, whose images were captured by a scanning electron microscope (SEM), following the same procedure adopted in Kim et al. (2020). Since data acquisition involved manual maneuvering of the microscope and defects can vary in size, these have introduced natural variations in image background and image/defect scales, adding challenges on top of standard defect detection. The acquired data can be categorized into five types of backgrounds, as illustrated in Fig. 5. Diagonal pattern refers to images with a repeating diagonal pattern, while Striped pattern includes images with a horizontal stripe pattern. Flat indicates plain images with no distinct patterns. Textured represents images with random textures on

Table 1

A list of defect detection datasets used to validate the effective of our proposed method. N denotes the number of normal samples, and A denotes the number of anomaly samples.

Dataset	Real	Resolution ($H \times W$)	# defect types	# normal img./# anomaly img.	# train (N/A) img./test (N/A) img.
SEM wafer	✓	480 × 480	10	3300/1535	4235 (3000/1235)/600 (300/300)
DTD-Synthetic (Aota et al., 2023)	X	300 × 300	5	300/300	300 (150/150)/300 (150/150)
MTD (Huang et al., 2020)	✓	222 ~ 394 × 103 ~ 620	5	952/392	1144 (852/292)/200 (100/100)

Table 2

Description of defect classes on SEM wafers. These defects represent the 10 most critical wafer defects essential for improving the wafer yield.

Class ID	Name	# imgs	Descriptions	Defect severity
1	BRG	208	This refers to a circuit short, appearing as foreign material adhered to the pattern.	“Yield killers” Defects that almost always induce a hard short or open
2	HBRG	35	Similar to the BRG defect type, but with only two lines are short-circuited.	
3	THNN	94	The pattern is thinned or broken, and samples corresponding to it are rare.	
4	BFBG	137	A bridge occurs due to causes in the previous layer process, characterized by a stain-like or donut-shaped pattern on the surface.	
5	GDE	43	A pattern or film has peeled off in a pit-like shape.	“High-risk” Defects that frequently cause failure but can be layout-dependent
6	LSAC	53	A pocket-shaped defect abnormally formed by a foreign object, such as a large particle or flake as a seed.	
7	RING	329	A pattern or film forms around the foreign object, creating a ring-like structure around it.	
8	CMSC	68	Characterized by a horseshoe shape, with single or multiple horseshoes appearing repeatedly in a row.	
9	SPOT	530	Caused by the formation of a thin film or pattern over a round seed.	“Process indicators” Defects that seldom cause functional failure
10	DESF	38	Similar to the SPOT defect type, but with a bumpy surface.	

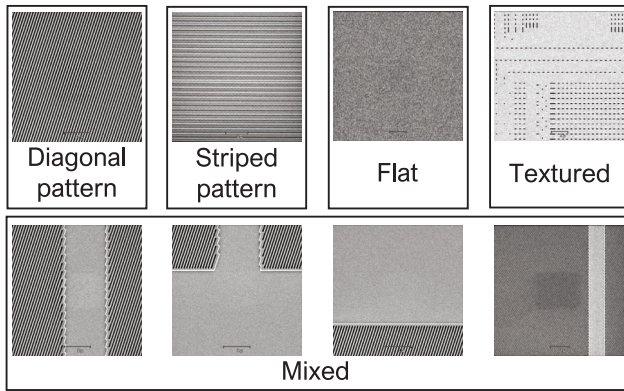


Fig. 5. Examples of images categorized into five background types in our SEM wafer dataset, namely, diagonal pattern, striped pattern, flat, textured and mixed. Note mixed comprises different geometric combinations of textures.

an otherwise plain background. Lastly, Mixed comprises images with flat regions and diagonal patterns randomly blended together. The acquisition scale of the images ranges from 0.2 μm to 2 μm , with defect sizes spanning approximately 0.1 μm to 4 μm .

SEM wafer includes 11 classes: one normal and ten defect classes with a long-tail distribution. The ten types of defects are shown in Fig. 6, where the number of data samples for each defect type is also indicated. Table 2 provides detailed descriptions of the ten types of defects. The dataset contains 3300 normal and 1535 defect images, all of which were reviewed by professionals. Also, the 10 defect classes

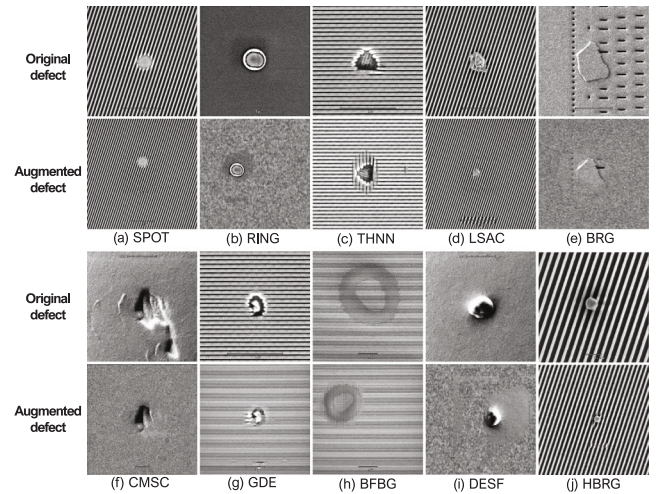


Fig. 6. Augmentation results of SegMix for 10 defect classes on SEM wafers. (Top) original defect images, (bottom) corresponding augmented images.

are distributed in a long-tailed manner, with the most populous class containing 530 images and the least populous only 35. For testing, we used 300 normal images and 300 defect images, with 30 images from each defect class. The ground truth for defect images in the test set was manually generated.

Public datasets. Additionally, we incorporated public datasets into our evaluation to further validate the effectiveness of our proposed

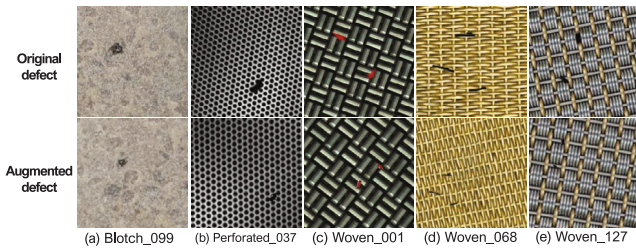


Fig. 7. Augmentation results of SegMix on 5 defect classes of DTD-synthetic (Aota et al., 2023). (Top) original defect images, (bottom) corresponding augmented images. The number of data for each defect type is: Blotch_099: 80, Perforated_037: 70, Woven_001: 60, Woven_068: 50, Woven_127: 40.

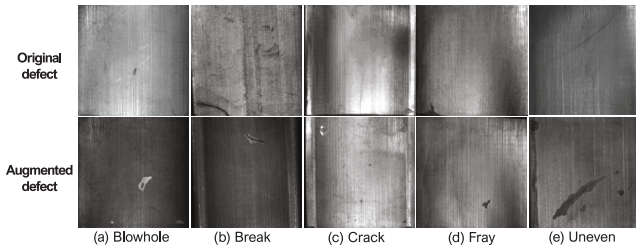


Fig. 8. Augmentation results of SegMix on 5 defect classes of MTD (Huang et al., 2020). (Top) original defect images, (bottom) corresponding augmented images. The number of data for each defect type is: Blowhole: 115, Break: 85, Crack: 57, Fray: 32, Uneven: 103.

method. Firstly, the DTD-Synthetic (Aota et al., 2023) dataset was introduced to address a variety of background conditions, encompassing five defect classes. In addition to utilizing the original dataset for defect detection analysis, we introduced multi-scale variations by applying random cropping to the original images (see Table 4). This dataset is structured with a long-tailed distribution of defects. Details and specific numbers of images per defect type for dataset can be found in Fig. 7. For testing, 150 normal and 150 defect images were selected, with 30 images per defect class.

The Magnetic Tile Defects (MTD) dataset (Huang et al., 2020), characterized by a long-tailed distribution, features a variety of scales and is based on real-world data. This dataset includes five types of defects, as depicted in Fig. 8, where the number of data samples for each defect type is also indicated. For testing, we selected 100 normal and 100 defect images, with each defect class represented by 20 images each.

4.2. Experimental settings

Evaluation metrics. Our experiments encompassed both defect detection and classification tasks. For anomaly detection, we used a binary classification approach, differentiating normal data from combined defect data, while for classification, we differentiated among all 10 defect classes. In contrast, the classifier we compared against includes an additional class for normal data, making it a total of 11 classes. We evaluated defect detection using both the *area under the receiver operating characteristic (AUROC)* and the *area under the precision-recall (AUPR)* curves. Our assessments included image-wise AUROC (I-AUROC) and AUPR (I-AUPR) for overall image classification, pixel-wise AUROC (P-AUROC) and AUPR (P-AUPR) for localizing anomalies, and specifically focused pixel-wise AUROC and AUPR for anomaly-only images (A-AUROC, A-AUPR) to precisely identify defects while excluding normal data. We followed the standard approach adopted in prior works (Defard et al., 2021; Roth et al., 2022), where the final anomaly score for the entire image is defined as the maximum

value of the anomaly map. Additionally, we conducted an ablation study on alternative global anomaly score aggregation strategies to assess their impact on overall detection performance, as presented in Table 17.

For estimating the misclassification rate during the detection or classification phase, we used the following equation:

$$\text{Misclassification rate} = \frac{FN + FP}{N + P}, \quad (17)$$

where N and P are the number of negative samples (normal) and positive samples (defects) respectively, FN (false negatives) specifically denotes the instances where defect classes are incorrectly classified as normal while, FP (false positives) denotes the number of normal instances misclassified as defects. We used the optimal F1 threshold to estimate FP and FN . For the classification task, the overall accuracy was measured using the top-1 accuracy across all classes including the normal instances. Furthermore, we categorized the training images into groups based on their quantity to assess performance across different data environments commonly used in long-tailed classifiers: over 100 images were labeled as “Many”, between 20 and 100 images as “Medium”, and fewer than 20 images as “Few”.

Implementation details. All experiments were carried out on a single NVIDIA A6000 GPU, Intel Xeon 2.80 GHz 64-core CPU with 512 GB DDR5 memory.

Based on the highest Silhouette Scores (Pedregosa et al., 2011), the SEM wafer data was divided into seven clusters, the DTD-Synthetic (Aota et al., 2023) into six, and the MTD (Huang et al., 2020) data into four. When training the classifier with the SegMix augmentation method, we applied an ‘over-balance’ (Shin et al., 2023a) approach to ensure each defect class would enjoy the benefit of augmented images. Specifically, we augmented the SEM wafer dataset classes to 1000 instances each, the DTD-synthetic dataset to 100 instances per class, and the MTD (Huang et al., 2020) dataset to 200 instances per class.

4.3. Defect detection results

Quantitative comparison. We compared our multi-cluster memory bank against commonly used models such as Patchcore (Roth et al., 2022) and PNI (Bae et al., 2023), models frequently used in different approaches, like PaDim (Defard et al., 2021) and SimpleNet (Liu et al., 2023). Additionally, we included reconstruction-based models like UniAD (You et al., 2022) and HVQ-trans (Lu et al., 2023b), which perform unified reconstruction across multiple classes. As shown in Table 3, the proposed method achieves the best performance on the SEM wafer dataset. Although there is minimal difference in I-AUROC and I-AUPR metrics, it is important to emphasize the significant improvements in pixel-wise metrics, particularly P-AUROC, P-AUPR, A-AUROC, and A-AUPR, where the method demonstrates notable gains. The model achieves superior performance across all metrics on the DTD-Synthetic (Aota et al., 2023) dataset, validating its effectiveness. On the MTD dataset (Huang et al., 2020), while the performance varies across models, our method consistently secures either first or second place (denoted with underline) in most metrics, further indicating its robustness. These results suggest that the proposed method’s focus on key defect regions significantly contributes to the high-quality augmented images produced by SegMix, which plays a crucial role in optimizing the detection threshold and improving classification performance.

Qualitative comparison. As shown in Fig. 9, both Patchcore (Roth et al., 2022) and PNI (Bae et al., 2023) fail to effectively handle diverse backgrounds and scales, often misclassifying non-defect regions as defects. As shown in these visualization results, this limitation is reflected in lower P/A-AUPR scores. While UniAD (You et al., 2022), shows some ability to distinguish between backgrounds, it does not perform as well as the proposed multi-cluster memory bank method, which more accurately segments only the defect regions. This precise

Table 3

Performance table showing the impact of applying the multi-bank approach in WaferDC, with I/P/A AUROC and I/P/A AUPR representing Imagewise, Pixelwise, and Anomaly-pixelwise AUROCs and AUPRs.

Methods	SEM wafer		DTD-Synthetic (Aota et al., 2023) (multi-scale)		MTD (Huang et al., 2020)	
	I/P/A AUROC	I/P/A AUPR	I/P/A AUROC	I/P/A AUPR	I/P/A AUROC	I/P/A AUPR
UniAD (You et al., 2022)	99.62/97.37/94.63	99.28/55.12/55.12	85.51/87.09/87.23	89.92/44.91/47.15	87.87/88.98/78.68	98.47/31.54/32.01
HVQ-trans (Lu et al., 2023b)	99.75/94.26/88.33	97.68/38.05/38.06	81.24/90.29/90.61	85.95/41.26/47.06	94.80/83.84/74.74	93.66/21.09/23.98
SimpleNet (Liu et al., 2023)	99.92/82.75/68.51	96.24/8.85/9.54	86.82/91.67/91.07	91.90/36.90/39.19	84.12/86.21/79.92	87.85/44.44/50.08
PaDim (Defard et al., 2021)	99.26/94.74/89.38	99.41/43.62/43.71	78.14/82.87/83.59	87.74/31.90/36.17	87.08/86.29/80.57	89.53/44.41/52.86
PNI (Bae et al., 2023)	99.79/81.93/66.39	99.80/8.48/8.60	88.09/90.83/90.06	90.78/22.93/27.88	97.34/91.93/84.77	97.67/52.82/53.10
Patchcore (Roth et al., 2022)	99.96/92.55/84.75	99.96/16.34/16.34	86.27/92.00/91.93	92.00/42.83/44.66	96.84/91.06/80.42	97.33/50.94/54.57
WaferDC (ours)	99.98/97.60/95.12	99.98/64.34/63.31	96.94/94.41/93.82	98.39/56.68/64.89	98.49/91.82/83.11	98.15/53.42/51.47

Table 4

Comparison of major defect detection methods on both the original and multi-scale customized DTD-Synthetic datasets.

Method	DTD-Synthetic (original)	DTD-Synthetic (multi-scale)
	I/P/A AUROC	I/P/A AUROC
UniAD (You et al., 2022)	89.71/90.10/89.35	85.51/87.09/87.23
Patchcore (Roth et al., 2022)	86.81/92.51/92.03	86.27/92.00/91.93
WaferDC (ours)	96.88/95.65/95.14	96.94/94.41/93.82

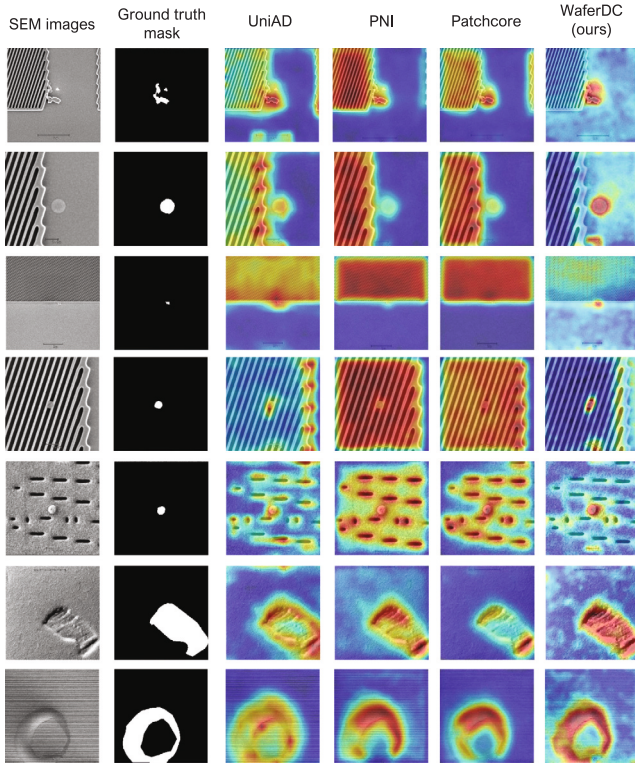


Fig. 9. Visualizations of defect segmentation maps. From left to right: SEM images, Ground truth mask and predicted anomaly heatmaps from UniAD, PNI, Patchcore and WaferDC (ours) respectively. Red indicates higher probability of being an anomaly, while blue represents a lower probability.

segmentation of defect regions enables higher-quality augmented images for SegMix by better identifying the defect segments. These visual results are shown in Figs. 6, 7 and 8 while the quantitative gains from using SegMix are presented in Section 4.5.

Robustness to multi-scale images. To address potential concerns about multi-scale customization of the DTD-Synthetic dataset, we also conducted experiments with the original untouched dataset. As shown in Table 4, existing models show a drop in performance on the multi-scale dataset, whereas our model consistently maintains high accuracy.

Visualization of clustered result of multi-cluster memory bank.

Fig. 10 illustrates the clustering results on the SEM Wafer dataset, showing that each cluster groups samples with similar background textures. Notably, complex overlapping textures, such as those in cluster 5 (approximately 50% striped regions) and cluster 7 (smaller striped regions), are successfully grouped into coherent clusters, highlighting the robustness and effectiveness of the proposed clustering approach in handling challenging visual patterns. This observation is further supported by the quantitative results using the IC-LPIPS (Ojha et al., 2021), as presented in Table 5, which show that intra-cluster diversity is reduced compared to the single memory bank setting.

Training runtime and inference speed. As shown in Table 6, we first evaluated the training and inference runtime, measured in seconds, corresponding to the construction of our multi-cluster memory bank. While the inclusion of additional clustering steps increases the construction time compared to a single memory bank approach, partitioning the memory bank into multiple clusters significantly reduces the number of comparisons required during nearest neighbor (NN) search. Since each individual cluster contains fewer features, this approach offers advantages in inference time compared to the single memory bank method. This also lowers memory consumption, ultimately resulting in faster inference. To ensure consistency in our comparison, we used the same base model equipped with the DINO pretrained encoder, while only varying the presence or absence of the multi-cluster memory bank. Moreover, since the multi-cluster memory bank compares the query image features with fewer samples sharing the same texture, we would anticipate the defect region to yield higher anomaly score compared to the standard memory bank. To confirm this empirically, we compared the nearest neighbor anomaly score for 2×2 pixel regions in defect and normal test images, both with and without the multi-cluster memory bank. Using the anomaly score, we calculated the average distances across all test images. The results reveal that while the anomaly score for normal pixels remains stable, the anomaly score for defect pixels increases, indicating better separation of defect regions when utilizing our multi-cluster memory bank strategy.

4.4. Defect classification results

We compared misclassification rate and top-1 accuracy of our method against other long tailed recognition methods (Zhang et al., 2023) on SEM wafer, DTD-Synthetic (Aota et al., 2023) and MTD (Huang et al., 2020) datasets in Table 7. Note the top-1 accuracy results incorporate the misclassified samples from defect detection for a fair comparison.

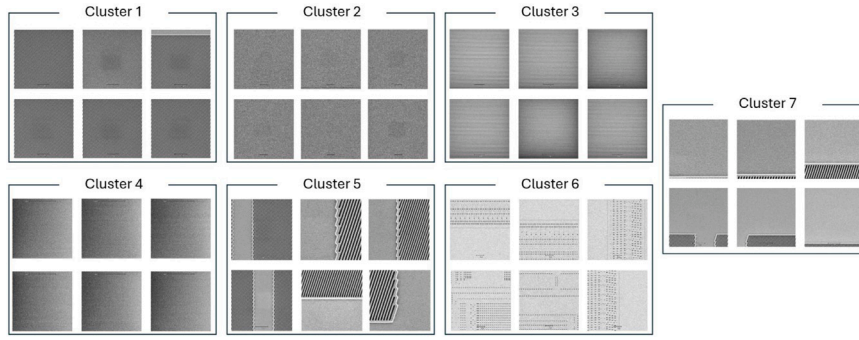


Fig. 10. Illustration of clustering results on the SEM Wafer dataset, where each cluster groups samples with similar background characteristics.

Table 5

Comparison of IC-LPIPS scores based on clustered images. The final IC-LPIPS is the average over all clusters for multi-cluster memory bank.

Method	IC-LPIPS		
	SEM wafer	DTD-Synthetic (Aota et al., 2023)	MTD (Huang et al., 2020)
Single memory bank (baseline)	0.319	0.296	0.322
Multi-cluster memory bank (ours)	0.238	0.216	0.258

Table 6

Comparison of training and inference runtime, inference speed (FPS), memory consumption, and anomaly scores (observed for defect and normal pixels) with and without the multi-cluster memory bank.

Methods	Training runtime (s)				Inference runtime (ms)			FPS (f)	Memory (MB)	Average anomaly score	
	PCA	K-means	Memory bank	SegMix	Memory selection	NN-search	Threshold-optm			Defect pixels (f)	Normal pixels (f)
Single memory bank (baseline)	–	–	34.28	–	–	63.73	0.49	15.57	7305	0.918	0.065
Multi-cluster memory bank (ours)	13.95	0.45	29.39	65.39	2.15	44.36	0.86	21.11	2951	0.938	0.065

Table 7

Performance comparison table showing misclassification and classification accuracy of WaferDC and other long-tail classifiers.

Type	Methods	FPS(↑)	SEM wafer				DTD-Synthetic (Aota et al., 2023) (multi-scale)					MTD (Huang et al., 2020)					
			Detection (%)		Classification (%)		Detection (%)		Classification (%)			Detection (%)		Classification (%)			
			Misclass. rate (↓)	Overall	Many	Medium	Few	Misclass. rate (↓)	Overall	Many	Medium	Few	Misclass. rate (↓)	Overall	Many	Medium	Few
Loss-based	Softmax (baseline)	84.20	4.3	76.0	90.0	70.0	63.3	38.7	15.9	9.9	25.0	0.0	50.0	50.0	100.0	0.0	0.0
	BS (Ren et al., 2020)	71.56	1.5	78.3	85.8	64.4	82.2	40.2	51.8	0.0	0.7	55.6	47.5	40.0	66.0	15.0	10.0
	LADE (Hong et al., 2021)	72.42	4.7	76.7	90.0	60.0	75.6	59.8	40.2	100.0	0.0	0.0	50.0	10.0	0.0	25.0	0.0
Ensemble-based	LDAM (Cao et al., 2019)	73.74	4.3	78.0	89.2	62.2	78.9	40.2	59.0	29.7	98.3	0.0	50.0	50.0	100.0	0.0	0.0
	RIDE (Wang et al., 2021)	52.08	3.7	80.0	86.7	72.2	78.9	40.2	59.8	0.0	100.0	100.0	37.5	56.5	<u>97.0</u>	11.3	35.0
	SADE (Zhang et al., 2022)	52.28	3.5	82.3	89.2	73.3	82.2	40.2	59.8	0.0	100.0	100.0	50.0	50.0	100.0	0.0	0.0
Self-supervised	GLMC (Du et al., 2023)	80.99	<u>1.0</u>	83.0	<u>93.3</u>	75.6	76.7	40.2	59.4	0.0	<u>98.9d</u>	100.0	50.0	10.0	0.0	0.0	99.9
PEFT based	PEL (Shi et al., 2023)	42.18	1.3	87.3	91.7	82.2	86.7	<u>7.2</u>	92.8	92.1	95.8	83.3	<u>6.5</u>	90.0	90.0	90.0	90.0
WaferDC (ours)		42.18	0.7	94.7	98.8	84.4	<u>85.6</u>	5.2	94.8	<u>94.1</u>	94.4	100.0	6.0	92.5	94.0	92.5	85.0

In a dataset with these characteristics, our model demonstrated high defect detection and classification performance, with a misclassification rate improved by 0.7% and overall top-1 accuracy reaching 94.7%, compared to existing methods.

For the DTD-Synthetic (Aota et al., 2023) and MTD (Huang et al., 2020) datasets, models based on CNNs (Zhang et al., 2023) and techniques from learning features across the entire image tend to confuse defect classes with the normal class, leading to lower detection and classification performance compared to ViT-based PEL (Shi et al., 2023) methods that learn features at the patch level.

We leveraged PEL (Shi et al., 2023) over other classifier models due to its robustness and efficiency. Softmax tends to overfit head classes due to its bias toward frequent labels. Balanced Softmax (BS) (Ren et al., 2020) mitigates this by reweighting logits based on label frequency, but its performance heavily relies on accurate priors and is incompatible with additional logit adjustments. LADE (Hong et al., 2021) further requires access to test label distributions, which are typically unavailable in real-world settings. Ensemble-based methods, including LDAM (Cao et al., 2019), RIDE (Wang et al., 2021), and SADE (Zhang et al., 2022), achieve improved performance through expert diversity or class-aware margins, but require careful scheduling and incur significant computational and memory overhead. GLMC (Du

et al., 2023) still relies on handcrafted mixing strategies and manual loss balancing, limiting its flexibility. In contrast, PEL (Shi et al., 2023) addresses these challenges by freezing the pre-trained backbone and adapting with lightweight modules that preserve generalizable representations, reduce overfitting on tail classes, eliminate reliance on external data or test priors, and achieve competitive or superior performance with lower training time and compute cost.

Moreover, compared to PEL (Shi et al., 2023) method, our WaferDC showed improved detection and classification performance. For the DTD-Synthetic (Aota et al., 2023) dataset, the misclassification rate decreased from 7.2% to 5.2%, while the overall top-1 accuracy improved from 92.8% to 94.8%. Similarly, in the MTD (Huang et al., 2020) dataset, the misclassification rate was decreased from 6.5% to 6.0%, and the overall top-1 accuracy increased from 90.0% to 93.0%. Although WaferDC has a slower FPS compared to other models, it still allows for time-efficient inspection, making it a practical solution for industrial applications.

Effect of data augmentation strategies on defect classification performance. To demonstrate the effectiveness of SegMix, we conducted a comparative evaluation against other data augmentation techniques, including MixUp and CutMix, as presented in Table 8. The results show that SegMix consistently achieves state-of-the-art performance in defect classification tasks. This improvement is attributed to

Table 8

Performance comparison table showing misclassification and classification accuracy of different data augmentation strategies.

Methods	SEM wafer					DTD-Synthetic (Aota et al., 2023) (multi-scale)					MTD (Huang et al., 2020)							
	Detection (%)		Classification (%)			Detection (%)		Classification (%)			Detection (%)		Classification (%)					
	Misclass. rate (↓)		Overall	Many	Medium	Few	Misclass. rate (↓)		Overall	Many	Medium	Few	Misclass. rate (↓)		Overall	Many	Medium	Few
Mixup	3.3		91.0	97.1	80.0	73.3	6.7		93.3	94.1	98.9	86.7	6.5		92.5	94.0	91.3	85.0
CutMix	3.3		91.0	97.1	77.8	75.6	5.7		94.3	90.7	98.9	100.0	8.0		90.5	94.0	91.3	70.0
SegMix (ours)	0.7		94.7	98.8	84.4	85.6	5.2		94.8	94.1	94.4	100.0	6.0		92.5	94.0	92.5	85.0

Table 9

Ablation study of individual proposed components in WaferDC in terms of detection and classification accuracies.

Methods	SEM wafer		DTD-Synthetic (Aota et al., 2023) (multi-scale)		MTD (Huang et al., 2020)	
	Detection (%)	Classification (%)	Detection (%)	Classification (%)	Detection (%)	Classification (%)
	(Misclass. rate (↓))	(Overall Top-1)	(Misclass. rate (↓))	(Overall Top-1)	(Misclass. rate (↓))	(Overall Top-1)
WaferDC (ours)	0.7	94.7	5.2	94.8	6.0	92.5
(w/o) multi-cluster memory bank	2.8	92.5	15.9	84.1	7.5	91.5
(w/o) SegMix, defect augmentation	2.8	89.2	6.8	93.2	9.0	88.5
(w/o) PEL (Shi et al., 2023) classifier, with softmax	0.7	79.7	5.2	49.4	6.0	46.0

the ability of SegMix that blends realistic defect regions in a structure-aware manner. In contrast, CutMix inserts arbitrary rectangular patches based solely on class-wise labels, often failing to preserve the spatial coherence and structural characteristics of real anomalies. Similarly, MixUp naively interpolates two entire image regions, which can blur semantic boundaries and lead to ambiguous samples. As a result, methods like CutMix and MixUp tend to increase the misclassification rate, whereas SegMix generates more semantically meaningful samples.

4.5. Ablation study

Effect of individual components in WaferDC. Table 9 reports the detection and classification performance with and without the key components in WaferDC, namely the multi-cluster memory bank and SegMix defect augmentation. Also, to verify the applicability of our methods across different classifiers, experiments were conducted using ResNext50 (Zhou et al., 2021) and PEL (Shi et al., 2023) within the classification module.

Firstly, implementing defect augmentation with a single memory bank improved overall top-1 accuracy over traditional PEL (Shi et al., 2023), though it increased the misclassification rate. Augmented data proved less effective in the standard single-bank setting, highlighting that it is more beneficial for threshold adjustment in a multi-cluster memory bank.

Second, using multiple memory banks without defect augmentation and relying solely on real validation data resulted in higher misclassification rates during defect detection, subsequently lowering defect classification performance. This drop in top-1 classification accuracy may also be attributed to the lack of augmented data for training. This highlights that augmented defect images not only boost classifier training but also reduce the misclassification rate in the detection module, demonstrating the synergy between the two modules.

Lastly, by replacing PEL (Shi et al., 2023) with a ResNext50-based (Zhou et al., 2021) baseline model (Softmax model in Table 7) in our proposed stratified network's classifier, we observe that, while the defect detection method remains the same and misclassification rates remain consistent, the overall performance is reduced due to the lower effectiveness of the baseline classifier model. This suggests that using the PEL classifier is a more effective choice for achieving better classification accuracy.

These experimental results indicate that the methods proposed in WaferDC achieve the highest performance improvements when used together, rather than individually.

Effect of dimensionality reduction methods. Dimensionality reduction methods were primarily employed to enable efficient clustering, as omitting them leads to an approximately sixteen-fold increase in clustering latency while maintaining comparable performance. Since the multi-cluster memory bank comprises PCA and K-means clustering methods, we conducted an ablation study to measure the impact of dimensionality reduction and clustering methods. As shown in Table

10, alternative dimensionality reduction methods, such as Independent Component Analysis (ICA), yield comparable performance, and no substantial performance gap is observed between the use of original and reduced features.

To further validate whether the reduced features preserve discriminative information, we conducted an ablation study to evaluate the percentage of standard deviation retained after applying PCA, relative to the original feature vectors, as shown in Table 11. The results confirm that reducing the feature dimensionality to 100 preserves sufficient variance for effective clustering, which is also supported by the t-SNE visualization in Fig. 11. Additionally, Table 12 shows that reducing the dimensionality to as few as 50 dimensions maintains anomaly detection performance without degradation.

Effect of clustering methods. Alternative clustering methodologies, such as K-medoids and DBSCAN, can also be utilized for feature clustering, and the results indicate that these methods yield relatively similar anomaly detection performance. In addition, well-established literature (Frigui and Krishnapuram, 2002; Lerasle et al., 2019; Tiwari et al., 2020; Acharya et al., 2025) demonstrates that medoid representations are more robust to outliers. Motivated by this, we conducted an experiment and found that while the choice between medoid and mean as the representative feature has a mild influence during the inference phase when selecting the cluster, it does not lead to significant differences in detection performance.

To conclude, our implementation employs PCA for dimensionality reduction, K-means for clustering, and the mean feature within each cluster as the representative. However, alternative approaches can also be utilized for constructing the multi-cluster memory bank, as the overall performance is not substantially impacted by these variations.

Effect of changing number of clusters in multi-cluster memory bank. Computing anomaly scores using all features from a single cluster exacerbates false-positive rates, since the greater number of pairwise feature comparisons increases the likelihood of spurious matches. As shown in Table 14, using a single cluster results in the lowest anomaly detection performance. In contrast, the multi-cluster memory bank exhibits consistent performance improvements across different numbers of clusters. Notably, the configuration with the highest silhouette score achieves performance that is competitive with state-of-the-art methods.

Effect of leveraging different ViT encoders for constructing the multi-cluster memory bank. We adopted DINO (Caron et al., 2021) as the ViT encoder for extracting feature vectors. Other ViT encoders, such as MAE ViT (He et al., 2022) and DeiT (Touvron et al., 2021), can also be integrated into our framework and demonstrate comparable performance to the DINO backbone, as shown in Table 13, highlighting that our framework is not sensitive to the choice of pretrained backbone network.

Effects of applying different blending techniques and transformations in SegMix. We investigated the impact of different blending techniques and found that our choice of alpha-blending strategy outperforms other methods, suggesting that alpha-blending produces more

Table 10

Performance comparison showing the impact of dimensionality reduction, clustering methods, and cluster representative feature selection on the construction and inference stages of the multi-cluster memory bank, with I/P/A AUROC and I/P/A AUPR representing Image-, Pixel-, and Anomaly-pixel-wise AUROCs and AUPRs.

Category	Methods	SEM wafer		DTD-Synthetic (Aota et al., 2023) (multi-scale)		MTD (Huang et al., 2020)	
		I/P/A AUROC	I/P/A AUPR	I/P/A AUROC	I/P/A AUPR	I/P/A AUROC	I/P/A AUPR
Reduction methods	No reduction	99.89/97.76/95.39	99.91/63.57/64.27	96.88/95.17/93.69	98.30/58.08/64.38	98.71/91.89/83.22	98.80/55.10/53.06
	ICA	99.60/98.00/95.91	99.53/63.42/61.65	96.55/94.48/92.82	97.81/55.02/58.16	95.71/89.42/79.95	96.17/54.30/54.73
Clustering methods	K-medoids	99.43/98.01/96.08	97.69/60.31/66.19	97.54/95.15/93.52	98.43/57.02/63.00	97.79/92.02/83.78	97.80/53.87/52.72
	DBSCAN	99.52/98.13/96.31	98.02/60.67/66.18	97.54/94.74/93.26	98.63/56.00/62.21	98.18/92.37/84.12	98.41/60.03/57.54
Representative feature	Medoid feature	99.88/97.62/95.10	99.91/62.92/63.56	95.65/95.23/93.80	97.02/57.61/64.34	98.38/90.14/82.55	98.36/55.55/53.31
WaferDC	PCA/K-means/Mean feature	99.98/97.68/95.22	99.97/63.18/63.62	96.85/95.23/93.80	98.67/57.61/64.34	98.19/91.99/83.45	98.09/55.34/53.13

Table 11

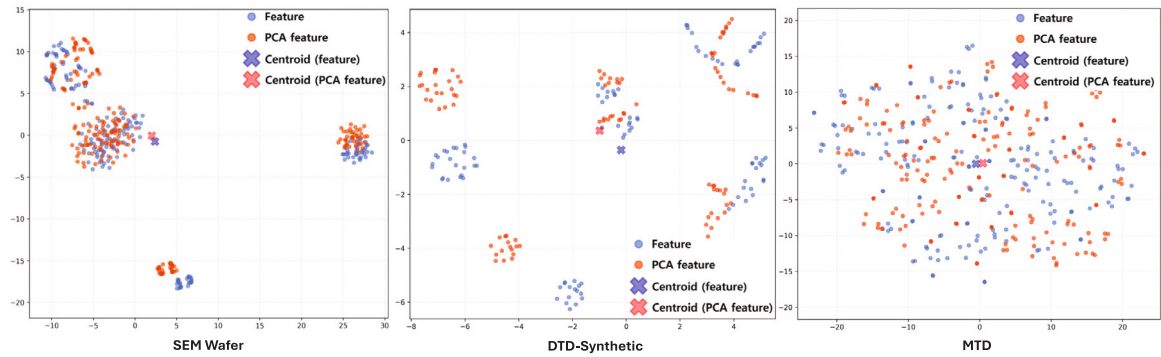
Ratio of preserved standard deviation ratio of feature vectors after reducing the dimensionality to 100 via principal component analysis (PCA).

	SEM wafer	DTD-Synthetic (Aota et al., 2023) (multi-scale)	MTD (Huang et al., 2020)
Ratio	0.963	0.942	0.887

Table 12

Performance table showing the impact of dimensionality reduction sizes on constructing the multi-cluster memory bank, with I/P/A AUROC and I/P/A AUPR representing Imagewise, Pixelwise, and Anomaly-pixelwise AUROCs and AUPRs.

Dimensions	SEM wafer		DTD-Synthetic (Aota et al., 2023) (multi-scale)		MTD (Huang et al., 2020)	
	I/P/A AUROC	I/P/A AUPR	I/P/A AUROC	I/P/A AUPR	I/P/A AUROC	I/P/A AUPR
50	99.85/97.85/95.57	99.89/61.73/63.43	96.85/95.29/93.84	98.67/57.73/64.99	98.40/91.98/83.40	98.19/55.22/53.02
100 (ours)	99.98/97.68/95.22	99.97/63.18/63.62	96.85/95.23/93.80	98.67/57.61/64.34	98.19/91.99/83.45	98.09/55.34/53.13
150	99.86/97.84/95.56	99.90/61.23/62.69	96.82/95.11/93.65	98.69/57.42/64.05	98.40/91.97/83.37	98.20/55.01/52.99

**Fig. 11.** The t-SNE visualization of original (blue) and PCA-reduced features (orange).**Table 13**

Performance table showing the impact of different ViT encoders on the construction of the multi-cluster memory bank, with I/P/A AUROC and I/P/A AUPR representing Imagewise, Pixelwise, and Anomaly-pixelwise AUROCs and AUPRs.

Encoders	SEM wafer		DTD-Synthetic (Aota et al., 2023) (multi-scale)		MTD (Huang et al., 2020)	
	I/P/A AUROC	I/P/A AUPR	I/P/A AUROC	I/P/A AUPR	I/P/A AUROC	I/P/A AUPR
MAE ViT	99.89/98.51/95.93	99.87/65.68/66.07	94.41/94.90/94.09	97.88/55.86/64.70	98.27/90.03/80.03	98.23/53.47/51.25
DeiT	99.81/98.03/95.95	99.72/60.27/61.08	95.97/95.42/94.45	97.54/56.41/66.03	98.62/91.93/83.40	99.01/53.41/52.40
DINO (ours)	99.98/97.68/95.22	99.97/63.18/63.62	96.85/95.23/93.80	98.67/57.61/64.34	98.19/91.99/83.45	98.09/55.34/53.13

Table 14

Performance comparison showing the impact of the number of clusters in the multi-cluster memory bank approach, where 7 clusters were used for the SEM Wafer dataset, and 6 clusters were selected for both DTD-Synthetic and MTD datasets, with I/P/A AUROC and I/P/A AUPR representing Imagewise, Pixelwise, and Anomaly-pixelwise AUROCs and AUPRs.

# of cluster	shilhouette score	SEM wafer		DTD-Synthetic (Aota et al., 2023) (multi-scale)		MTD (Huang et al., 2020)	
		Wafer/DTD-Syn/MTD	I/P/A AUROC	I/P/A AUPR	I/P/A AUROC	I/P/A AUPR	I/P/A AUPR
1	- / - / -		99.37/94.80/89.33	99.54/36.53/36.53	91.69/90.24/86.65	95.19/29.29/29.67	98.40/49.90/50.14
2	0.360/0.184/0.095		99.93/97.43/94.75	99.94/61.38/61.32	91.94/90.86/88.01	95.30/43.86/52.45	98.78/91.37/82.54
3	0.454/0.266/0.100		99.90/98.27/96.44	99.87/66.49/66.60	94.81/94.09/92.90	96.69/50.54/59.15	98.30/56.00/54.95
4	0.468/0.301/0.110		99.93/98.29/96.45	99.92/66.20/66.97	95.13/94.99/93.47	96.74/55.68/59.87	98.80/89.91/79.80
5	0.516/0.292/0.103		99.95/97.35/95.55	99.95/64.74/64.85	95.82/95.17/93.87	97.25/59.32/63.53	98.45/91.55/82.49
6	0.554/0.322/0.114		99.85/96.60/95.06	99.91/61.08/62.90	96.85/95.23/93.80	98.67/57.61/64.34	98.19/91.99/83.45
7	0.575/0.320/0.111		99.98/97.68/95.22	99.97/63.18/63.62	96.85/95.01/93.76	96.51/56.69/63.34	99.07/90.56/81.14
8	0.510/0.278/0.113		99.86/96.86/94.70	99.91/60.29/60.79	91.49/95.15/93.85	97.35/61.12/64.13	98.37/91.22/82.12
9	0.474/0.213/0.113		99.91/97.06/94.01	99.96/59.47/60.86	90.47/94.69/93.32	96.38/56.95/64.31	98.80/89.41/79.60
10	0.465/0.205/0.113		99.75/97.31/94.53	99.87/61.52/61.54	90.92/94.93/93.54	96.52/58.01/64.15	98.39/89.99/80.72

Table 15

Performance comparison table showing misclassification and classification accuracy of different blending techniques.

Methods	SEM wafer					DTD-Synthetic (Aota et al., 2023) (multi-scale)					MTD (Huang et al., 2020)				
	Detection (%)	Classification (%)				Detection (%)	Classification (%)				Detection (%)	Classification (%)			
	Misclass. rate (↓)	Overall	Many	Medium	Few	Misclass. rate (↓)	Overall	Many	Medium	Few	Misclass. rate (↓)	Overall	Many	Medium	Few
SegMix + gaussian smoothing	2.7	88.7	93.3	86.7	84.4	5.7	94.3	92.0	98.9	96.7	10.0	88.5	94.0	90.0	55.0
SegMix + direct patch blending	1.3	90.0	93.3	92.2	83.3	6.7	93.3	94.1	98.9	86.7	7.0	91.5	94.0	91.3	80.0
SegMix + alpha blending (ours)	0.7	94.7	98.8	84.4	85.6	5.2	94.8	94.1	94.4	100.0	6.0	92.5	94.0	92.5	85.0

Table 16

Performance comparison of SegMix with and without rotation and scale augmentations, showing misclassification rates and classification accuracy. Rot. denotes rotation and S. denotes scaling.

Methods	IC-LPIPS	SEM wafer					IC-LPIPS	DTD-Synthetic					IC-LPIPS	MTD						
		Detection (%)		Classification (%)				Detection (%)		Classification (%)				Detection (%)		Classification (%)				
		Misclass. rate (↓)		Overall	Many	Medium	Few		Misclass. rate (↓)		Overall	Many	Medium	Few		Misclass. rate (↓)		Overall	Many	Medium
SegMix without Rot./S.	0.257	0.8		93.8	98.1	83.3	83.3	0.165	6.0		94.0	94.1	97.8	88.3	0.374		91.0	94.0	90.0	80.0
SegMix with Rot./S.	0.297	0.7		94.7	98.8	84.4	85.6	0.176	5.2		94.8	94.1	94.4	100.0	0.425	6.0	92.5	94.0	92.5	85.0

Table 17

Performance table showing the impact of different anomaly score aggregation, with I AUROC and I AUPR representing Imagewise AUROCs and AUPRs.

Methods	SEM wafer		DTD-Synthetic (multi-scale)		MTD	
	I-AUROC	I-AUPR	I-AUROC	I-AUPR	I-AUROC	I-AUPR
Softmax	99.98	99.96	96.85	98.67	98.23	98.11
Percentile-based pooling	99.93	99.91	95.65	97.02	98.38	98.35
Max pooling (baseline)	99.98	99.97	96.88	98.71	98.19	98.09

visually natural synthetic anomalies, as demonstrated in Table 15. Finally, Table 16 demonstrates improved feature diversity resulting from the transformations (rotation and scaling) applied during SegMix, as evidenced by the increased IC-LPIPS scores. Applying transformations also leads to enhanced misclassification and classification performance compared to SegMix without transformations.

Effect of applying different anomaly score-aggregation strategies. While we adopted defining the global anomaly score as the maximum patch anomaly score across all patches to ensure fair and consistent comparison with existing methods, we have conducted an ablation study as shown in Table 17. Overall anomaly detection performance remains relatively consistent across all aggregation methods.

Effect of setting different threshold values for defect mask generation in SegMix. We empirically observed that the threshold defined by $\mu + 2\sigma$ closely aligns with the midpoint of the anomaly score range, $(s_{\min} + s_{\max})/2$, and the region around $(s_{\min} + s_{\max})/2 - 3\sigma$ captures the remaining non-distinct anomalous regions, which have effectively blended into generated pseudo-anomaly images using SegMix. Thus, selecting high-confidence anomaly regions, while incorporating larger portions of non-distinct areas using an alpha-blending strategy (Foley, 1996), leads to improved performance as demonstrated in Table 18. Furthermore, we observed that our choice of upper and lower bound thresholds consistently yields the second-highest (red) or third-highest (blue) IC-LPIPS scores (Ojha et al., 2021), and we emphasize that the IC-LPIPS score primarily measures diversity. As a result, augmented defect images that do not closely resemble real defects may still achieve high IC-LPIPS scores. Therefore, a high IC-LPIPS score does not necessarily correspond to improved classification performance. While the choice of threshold can influence accuracy, our method continues to achieve state-of-the-art results when compared to existing classifiers, as shown in Table 7, although it does not yield the absolute highest performance. This suggests that the threshold selection does not significantly compromise overall accuracy or the effectiveness of the proposed data augmentation approach. Based on these observations, we have chosen the upper and lower bound threshold values for SegMix as $(s_{\min} + s_{\max})/2$ and $(s_{\min} + s_{\max})/2 - 3\sigma$, respectively.

5. Conclusion and future work

Our work addresses an important industrial challenge of accurately detecting and classifying wafer defects from scanned electron microscope images, which involves overcoming complex issues such as diverse background types, varying image or defect scales, and highly imbalanced defect distributions with long-tail characteristics. To tackle these challenges, we have presented WaferDC, a framework that employs a multi-cluster memory bank strategy to effectively handle variations in background and scale. Additionally, we have introduced SegMix, a novel defect augmentation technique tailored to counteract the impact of long-tail distributions, significantly enhancing defect detection and classification accuracy. Through extensive experiments across our proprietary SEM wafer dataset and public defect datasets, we have demonstrated overall improved defect detection and classification accuracies over the existing algorithms, with each of our proposed component contributing to enhanced performance. Through this work, we aim to establish a new benchmark for real-world defect detection and classification systems in semiconductor manufacturing, setting a higher standard in precision and reliability.

For future work, introducing theoretical rigor into the clustering formulation would strengthen the foundation of the proposed multi-cluster memory bank. Additionally, developing a more principled statistical framework for SegMix could improve its ability to handle outliers and skewed anomaly score distributions.

Limitations We list potential limitations of our proposed method. First, the effectiveness of WaferDC relies on the initial performance of the defect detection module. If the defect detection module is overly inaccurate or performs poorly, it can lead to decreased augmentation quality and consequently poorer detection and classification accuracies. Second, even though our method aims to naturally segment and cut out ambiguous defect areas, it can inadvertently include parts of the background in the defect images which can result in lower quality augmentations for background. Also, SegMix is dependent on hyperparameter choices when retrieving the predicted defect regions. Lastly, the assumption that PCA-reduced features preserve discriminative information is supported only by empirical evidence but not theoretically bounded. These issues necessitates further investigations into defect localization and augmentation which we leave for future work.

Table 18
Performance comparison of SegMix threshold in terms of classification accuracy and misclassification rate. For the IC-LPIS scores, red indicates the second-highest values, and blue indicates the third-highest values.

Method	Upper bound	Lower bound	SEM wafer					DTD-Synthetic Aota et al. (2023) (multi-scale)					MTD Huang et al. (2020)							
			Detection (%)	Classification (%)				Detection (%)	Classification (%)				Detection (%)	Classification (%)						
			Misclass. rate (↓)	Overall	Many	Medium	Few	IC-LPIS	Misclass. rate (↓)	Overall	Many	Medium	Few	IC-LPIS	Misclass. rate (↓)	Overall	Many	Medium	Few	IC-LPIS
SegMix	$\mu+1.0\sigma$	μ	1.1	91.7	95.2	84.4	82.2	0.387	6.3	93.3	94.1	94.4	93.3	0.315	7.5	90.5	94.0	88.0	80.0	0.466
	$\mu+1.5\sigma$	μ	0.8	94.0	96.7	86.7	88.9	0.360	6.3	93.3	94.1	94.4	93.3	0.286	7.0	91.0	94.0	88.8	85.0	0.436
	$\mu+2.0\sigma$	μ	0.9	93.0	96.9	81.1	86.7	0.362	6.3	93.3	94.1	94.4	93.3	0.301	6.5	92.0	94.0	90.0	90.0	0.411
	$(s_{\min}+s_{\max})/2$	μ	1.1	91.5	96.9	78.9	78.9	0.376	7.0	93.0	93.3	96.7	86.7	0.311	7.0	91.5	94.0	90.0	85.0	0.483
	$\mu+1.0\sigma$	$(s_{\min}+s_{\max})/2-3\sigma$	0.9	92.8	97.4	82.2	82.2	0.408	6.7	93.3	94.1	96.7	86.7	0.318	7.0	91.5	94.0	91.3	80.0	0.494
	$\mu+1.5\sigma$	$(s_{\min}+s_{\max})/2-3\sigma$	0.9	93.3	97.6	82.2	84.4	0.368	6.3	93.7	94.1	96.7	88.3	0.314	7.0	91.0	94.0	92.0	85.0	0.436
	$\mu+2.0\sigma$	$(s_{\min}+s_{\max})/2-3\sigma$	0.9	93.3	97.6	82.2	84.4	0.369	5.7	94.3	93.3	94.4	96.7	0.297	6.5	92.0	93.0	92.5	85.0	0.421
	$(s_{\min}+s_{\max})/2$	$(s_{\min}+s_{\max})/2-3\sigma$	0.7	94.7	98.8	84.4	85.6	0.377	5.2	94.8	94.1	94.4	100.0	0.317	6.0	92.5	94.0	92.5	85.0	0.482

CRediT authorship contribution statement

Taekyeong Park: Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yongho Son:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Investigation, Formal analysis, Data curation, Conceptualization. **Sanghyuk Moon:** Writing – original draft, Software, Resources, Methodology, Investigation, Data curation. **Seungju Han:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization. **Je Hyeong Hong:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Je Hyeong Hong, Sanghyuk Moon, Hyeon Jeong Park, Seungju Han has patent #18/357,475 pending to Samsung Electronics, Co., Ltd. Je Hyeong Hong, Taekyeong park, Sanghyuk Moon, Seungju Han has patent #KR 10-2024-0048252 and #US 19/056,033 pending to Samsung Electronics, Co., Ltd. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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